

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Requirements Table – Automated Rubric Assignment Evaluator

Student, Faculty Evaluator, Peer Reviewer

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall automatically queue uploaded student project zip files, execute pre-configured unit test suites, and generate an itemized rubric score breakdown.	High	Pass: Rubric breakdown is rendered with test results within 10 seconds of submission. Fail: File queue stalls or invalid rubric tally.	Core automated evaluation function; removes manual grading bottleneck.
FR-002	Functional	The system shall allow a Faculty Evaluator to define a rubric template (criteria names, weights, and maximum points per criterion) before an assignment's submission window opens.	High	Pass: Saved rubric displays all criteria with weights summing to 100%. Fail: Assignment opens for submissions while weights do not sum to 100%.	Ensures a consistent, unambiguous scoring baseline for automated evaluation.
FR-003	Functional	The system shall automatically assign each successfully tested submission to two randomly selected, anonymized peer reviewers and open a peer-review form for each reviewer.	Medium	Pass: Within 5 minutes of test completion, two reviewer assignments appear on reviewers' dashboards and reviewer identity is hidden from the submitter. Fail: Assignment stalls beyond 5 minutes or identity is exposed.	Eliminates manual peer-review distribution overhead described in the problem statement, while preserving fairness via anonymity.
FR-004	Functional	The system shall let a Student view their itemized rubric score breakdown, automated test results, and received peer review comments on a single dashboard page.	Medium	Pass: Dashboard loads and displays all three components within 3 seconds of login. Fail: Any component is missing or the dashboard fails to render.	Gives students transparent, timely feedback without requiring faculty follow-up.
FR-005	Functional	The system shall allow a Faculty Evaluator to manually adjust an automatically generated rubric score for any criterion, requiring a mandatory text justification before the change is saved.	Medium	Pass: An override attempt without justification text is rejected; a valid override updates the score and logs the original value, new value, and justification. Fail: Override saves with empty justification or without an audit log entry.	Automated scoring can misjudge edge cases (e.g., partial credit); faculty need an auditable manual override path.
NFR-001	Nonfunctional	[Type: Performance & Security] The system shall handle up to 100 concurrent project submissions without dropping queue items or exceeding 1.5 GB memory footprint.	High	Pass: Under a simulated load of 100 concurrent project submissions, no queue item is lost and memory usage remains at or below 1.5 GB. Fail: Any queue item is lost or memory usage exceeds 1.5 GB.	Guarantees stability during high-traffic periods such as assignment deadlines.
NFR-002	Nonfunctional	[Type: Security & Reliability] The system shall retain all submission files, test logs, and rubric scores for a minimum of 2 academic years and encrypt all stored score and personal data at rest using AES-256.	High	Pass: A data audit after 2 years confirms records are intact and readable only via authorized decryption. Fail: Records are purged early or stored data is found unencrypted.	Meets academic record-keeping obligations and protects sensitive student performance data.